

1) Single Topology

a. Command :-

- i. `sudo su`
- ii. `mn --topo single,3`

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Sep 24 11:31
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ mn --version
2.3.0
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ sudo su
[sudo: authenticate] Password:
root@sagnik-bachhar-VirtualBox:~# mn --topo single,3
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 h3 s1
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
h3-eth0<->s1-eth3 (OK OK)
mininet> net
h1 h1-eth0:s1-eth1
```

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Sep 24 11:32
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

h2-eth0<->s1-eth2 (OK OK)
h3-eth0<->s1-eth3 (OK OK)
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=6279>
<Host h2: h2-eth0:10.0.0.2 pid=6281>
<Host h3: h3-eth0:10.0.0.3 pid=6283>
<OVSBridge s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=6288>
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.56 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.046 ms
^C
--- 10.0.0.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6132ms
rtt min/avg/max/mdev = 0.046/0.412/2.558/0.875 ms
```

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Sep 24 11:32
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-ba: root@sagnik-bachhar-V x root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu
Files 0.0.0.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6132ms
rtt min/avg/max/mdev = 0.046/0.412/2.558/0.875 ms
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::a4ef:26ff:fe1f:a814 prefixlen 64 scopeid 0x20<link>
    ether a6:ef:26:1f:a8:14 txqueuelen 1000 (Ethernet)
    RX packets 63 bytes 6084 (6.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 27 bytes 2154 (2.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet> h1 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: h1-eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether a6:ef:26:1f:a8:14 brd ff:ff:ff:ff:ff:ff link-netnsid 0
```

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Sep 24 11:32
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-ba: root@sagnik-bachhar-V x root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: h1-eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether a6:ef:26:1f:a8:14 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.0.0.1/8 brd 10.255.255.255 scope global h1-eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a4ef:26ff:fe1f:a814/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
mininet> h2 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: h2-eth0@if8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 06:58:66:47:de:4a brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.0.0.2/8 brd 10.255.255.255 scope global h2-eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::458:66ff:fe47:de4a/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
mininet> h3 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: h3-eth0@if9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 72:2a:09:7b:c5:17 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.0.0.3/8 brd 10.255.255.255 scope global h3-eth0
        valid_lft forever preferred_lft forever
```

```
Sep 24 11:32
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar:~$ ifconfig h3-eth0 10.0.0.3
2: h3-eth0@if9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 72:2a:09:7b:c5:17 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.0.0.3/8 brd 10.255.255.255 scope global h3-eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::702a:9ff:fe7b:c517/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

mininet> h1 route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
10.0.0.0 0.0.0.0 255.0.0.0 U 0 0 0 h1-eth0
mininet> h2 route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
10.0.0.0 0.0.0.0 255.0.0.0 U 0 0 0 h2-eth0
mininet> h3 route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
10.0.0.0 0.0.0.0 255.0.0.0 U 0 0 0 h3-eth0
mininet> h1 ip route
10.0.0.0/8 dev h1-eth0 proto kernel scope link src 10.0.0.1
mininet> h2 ip route
10.0.0.0/8 dev h2-eth0 proto kernel scope link src 10.0.0.2
mininet> h3 ip route
10.0.0.0/8 dev h3-eth0 proto kernel scope link src 10.0.0.3
mininet> exit
*** Stopping 0 controllers
...
*** Stopping 3 links
...
*** Stopping 1 switches
s1
```

```
Sep 24 11:32
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar:~$ sudo mn -c
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwte
st mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpb
wtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
*** Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([_[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
```

```
Sep 24 11:33
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-ba: root@sagnik-bachhar-V x root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu root@sagnik-bachhar-Virtu

*** Stopping 3 hosts
h1 h2 h3
*** Done
completed in 1918.204 seconds
root@sagnik-bachhar-VirtualBox:/home/sagnik-bachhar# sudo mn -c
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwte
st mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpb
wtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
*** Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([_.:alnum:])+eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
root@sagnik-bachhar-VirtualBox:/home/sagnik-bachhar#
```